

"Digital Sanctums of Empowerment": Exploring Community and Everyday Resilience-Building Tactics in Online Professional Communities for Women

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ABSTRACT

Work and learning are essential facets of human life and legacy, yet sociocultural barriers may limit access to them. Historically, women have faced multiple restrictions that have hindered their entry into professional ventures. These constraints are even more palpable in technical domains, due to the added sociocultural barriers in these fields which have relegated women's voices to the margins. Through a mixed-methods comparative investigation of three professional development communities on Reddit - two focused on women, and one general professional online community serving as the baseline, we report and discuss the resilience-building, empowerment, and knowledge-development tactics that drive collective identity formation in these subreddits. The long-term goal of this work is to address the way in which such channels can be designed, maintained, and curated to offer spaces for enrichment, empowerment, and advocacy with a focus on professional development for women in technical domains.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

KEYWORDS

Resilience building and support; Online community norms and practice; Gender in the workplace; Content analysis; Network analysis; Reddit

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1 STUDY MOTIVATION AND RELATED WORK

Despite the powerful civil and women's rights movements of the 20th century, the ongoing struggle for equality in non-Western and Western contexts, and growing justice-oriented work in the academy today, issues of sexism and gender-based discrimination continue to operate in many organizational settings, including the field of information and technology (IT) [9, 14, 16]. Understanding and dismantling these oppressive practices is a complex process because gender has been shown to be not simply a fixed label of identity but rather a dynamic practice, shaped by, and in turn shaping, society and social systems [5, 31]. Essential to this conceptualization of gender is Pierre Bourdieu's Habitus framework which posits that sociocultural norms create an invisible, often symbolic, tradition of practice that may construe one's social position and in turn impact one's disposition. The gendered habitus adds further nuance to how a habitus of oppression and domination may inhibit, impair, and limit the extent of expression afforded to women across multiple contexts [3]. It is therefore not surprising that calls for developing strategies of endurance when faced with such adversities in the workplace have gained momentum and vital significance in feminist IT scholarship [9, 19]. This literature has often extended prior social constructivist work such as Bourdieu's, as well as career development and human-computer interaction (HCI) scholarship which has highlighted the growing power of online collectives to develop knowledge and social capital [29, 30]. Further, recent scholarship inspired by feminist HCI has highlighted how women-led communities foster practices guided by the values of advocacy, accessibility, and equitable participation [11]. Moreover, research shows that online communities have become indispensable conduits of support and care in times of crisis [1], especially when faced with deep-seated chronic crises such as patriarchal and sexist regimes [15, 17, 34]. This study builds on this powerful HCI framework for empowering marginalized voices and seeks to understand how online channels facilitate community-building and can act as safe havens for learning to navigate and actively resist the gendered habitus manifesting in the IT workplace.

In particular, we seek to address the following research questions:

- (1) *What forms of collective practices emerge in the women-centric professional online communities compared to the general, seemingly non-gendered, IT community, as indicated by their underlying content themes and social network structures?*

- (2) *Are there differences in the support- and resilience-building mechanisms across these communities, as demonstrated by the communal practices in each forum?*

2 METHOD

To address the central research questions of the study, we examine three Reddit communities at the intersection of gender, professional development, and IT: *cscareerquestions*, *careerwomen*, and *girls-gonewired* (selected due to their overlap in orientation with that of this investigation's focus).

2.1 Data collection

The dataset was compiled using a combination of existing databases on Reddit data (PSAW) and Python API wrappers (PRAW) for each of the three subreddits [35]. For this initial investigation a random sample was selected for each of the forums inspected. However, on manual inspection it was found that churn rates (in terms of number of threads and number of responses to threads) differ across the three forums of interest. Thus, the number of threads sampled from each forum was selected such that in aggregate, the total number of conversational units (each individual digital trace - a question or comment) were comparably similar across all the three datasets. Following best practices in Reddit research, all conversational threads that were removed or deleted were excluded from analysis [20]. Further descriptive and aggregate statistics are provided in Table 1.

2.2 Analysis

The analysis is based on a mixed-methods research design combining content analysis and social network analysis (SNA) [18]. This approach allows us to develop a more granular understanding of the typology and habitus of gendered practices among members of the three virtual communities, and unpack the interactions between the social positions and dispositions of community members [3]. We then validate the results through an econometric model to establish the significance of the findings [33].

Since textual conversations form the heart of the discourse patterns in these online channels, content analysis allows us to explore the types of underlying conversational themes that shape the community dynamics of these spaces [28]. To derive the content labels, a thematic coding process rooted in grounded theory was followed [6]. First, an initial set of ten exploratory themes were developed. Next, these themes were further refined, reframed, arriving at the final schema. Both authors discussed and validated the final schema comprising of six themes, mentioned in Section 3.1.

While content analysis helps capture the disposition of community members based on their conversational traces, social network analysis (SNA) helps establish the social structure and interaction patterns within each community by representing each community member as a node whose connections to other members, or edges, denote comments issued or received. Theoretically, this approach is helpful in understanding how positions or roles assumed by individuals and communities alike impact patterns of participation and communication. This type of analysis allows us to survey and compare each community in terms of community-building, resilience, and social support [2].

3 RESULTS

This section summarizes our findings from the content and network analysis, and forms the basis of a comparative understanding of the three forums investigated. The numerical insights are tabulated in tables 1, 2.

3.1 Description of content categories

Survival tactics & Empowerment: Posts in these category aim to share tacit skills and tactics about navigating implicit organizational expectations, such as prescriptions of conduct. Such posts also help facilitate greater social integration with others in the workplace. In the context of *r/cw* and *r/ggw*, this theme primarily depicts how experiential wisdom can help tackle the combined effect of the organizational and gendered habitus. By contrast, among *r/csc* members, this theme is more generic and aimed at self-development strategies that do not necessarily involve gendered tensions. For example, in *r/csc*, a conversation related to this topic detailed strategies to improve communication with coworkers about work commitments: *Whatever you care about most, you should ask about. If being able to work from home .. then you should find out if that's possible now..* Conversely, a conversation under this category in *r/cw* entailed inspiring and motivating other community members to not lose hope in tackling gendered exclusions: *On days when it is extra hard being a career women, I think of how hard it was for the women before us .. Keep doing what you're doing, we will all be an inspiration to someone one day.*

Resource development: These posts provide more instrumental and practical guidance centered on resources for growing one's career and network, and develop expertise in technical domains. Posts in this category also include information about training camps and online materials for self-improvement. Thus, this category highlights the knowledge capital and professional expertise offered by the community. An example conversation from *r/ggw* inquiring about remote training resources to sharpen one's technical ability states: *I'm searching bootcamps and if anyone can share their experience of one that is online?.*

Reflections on ethics: Conversations in this rubric question, contest, and critique the organizational ethos. Such posts highlight how online communities emerge as collectives of collaborative purpose and scaffold conversations that advocate for justice in the workplace. This theme illustrates a key distinction between the women-centric and general IT forums. For example, in *r/csc*, this category entails discussions pertaining to how the technical field operates. On the other hand, in *r/ggw*, this theme captures the struggles that underrepresented groups may face in tackling the demands of the technical workforce regime. Similarly, albeit more generally, conversations under this theme in *r/cw* discuss issues such as wage disparity, culture, and understanding gender-based exclusions. An example conversation from *r/csc* states: *Even an average data scientist should come to the table with data and visualization to back up their assertions* on a discussion regarding the expectations and necessary skill sets organizations need from data science professionals. In contrast, a post from *r/ggw* sharply criticizing the focus of training resources for women in technology states *The few female bootcamps here need to focus more hard skills, the tools, and field of interest and not just the issue of diversity.* Thus,

these discussions can serve as a provocative arena for debating how diversity should be embraced, performed, and maintained in the workplace.

Community preservation: Posts associated with this category aim at generating, sustaining, and upholding community values and beliefs. By including posts of affirmation and appreciation, this theme serves as an attempt to foster bonds through humor and philosophical discussions related to the workplace (particularly in conversations on r/ggw & r/cw). This shows a stronger thrust to community formation, sustenance, and maintenance. An example from r/cw states: *Thanks, I admire of the work you are doing, congrats*. This post illustrates appreciation for the way a community member contributes to the community and inspires others.

Navigating toxicity: Posts related to this theme include disclosures around the obstacles community members face as a result of adverse workplace culture, which can be seen as an example of a habitus of domination. Particularly, in the context of the women-centric forums (r/cw & r/ggw), these posts typically depict struggles with self-belief, and aim to capture the invisible strains induced by the symbolic violence that may exist in these technical domains [24]. An example post under this category from r/cw states: *As a woman I want to power through and fight the system, but I don't know if I want to put myself through that*, indicating the trauma induced by the symbolic forms of oppression and forces of exclusion that operate in workplaces.

Impression management: Every habitus constructs and imposes a behavioral schema that characterizes one's disposition and identity [21]. Thus, this category chiefly exemplifies the struggle on the path towards crafting one's voice and self-worth when aligning one's value system with organizational expectations [22]. An example conversation from r/cw states: *I do find it difficult to be assertive when it conflicts with my desire to be nice ... it is very difficult to do things that would make me seem rude*.

3.2 Network inferences

SNA-derived social network metrics provide a deeper sense of how community emerges in three online forums. Specifically, they offer a more micro-level view of these collectives through the study of dyadic interactions [23]. Key insights reveal that both r/ggw & r/cw have a large main component encapsulating 77% and 57% of the nodes respectively, whereas r/csc's largest component comprises only 15% of the nodes, indicating a stronger possibility of community formation in the women-centric subreddits. This is further supported by the observation that as compared to r/csc, both r/cw & r/ggw have higher eccentricity ($p=0.003$, $p=0.001$), greater closeness centrality ($p=0.003$, $p=0.02$), greater betweenness centrality ($p=0.04$, $p=0.02$) and lower eigenvector centrality ($p=0.04$, $p=0.001$). These centrality measures show that in the case of r/ggw & r/cw, a denser network emerges, whereas r/csc's network structure is fragmented and comprised of many smaller cliques, indicting a lack of community orientation. The stronger collective in the women-centric forums can be essential for social support, resilience and help in developing a unified voice and identity among community members [25]. Further, comparing r/ggw and r/cw reveals that r/cw has a lower indegree, eccentricity and clustering coefficient ($p=0.01$, $p=0.006$, $p=0.02$). Additionally, the greater number of connected

Table 1: Data stats & content inferences

Data statistic	r/ggw	r/cw	r/csc
Number of conversations	367	397	269
Earliest creation date	02/28/18	02/11/18	02/05/18
Latest creation date	10/15/21	12/30/20	12/23/20
Number of unique users	198	229	165
Content categories	(%)	(%)	(%)
Survival & Empowerment	34	32	27
Resource development	20	22	38
Community preservation	14	23	12
Navigating toxicity	8	7	6
Reflections on ethics	16	10	11
Impression management	8	6	6

Table 2: Comparative network inferences

Network insights	r/ggw	r/cw	r/csc
Number of nodes	195	170	158
Number of edges	253	192	173
Average degree	1.30	1.13	1.10
Network diameter	9	7	4
Connected components	10	23	23
Average clustering coefficient	0.05	0.01	0.02
Average eccentricity	2.45	1.82	1.39
Average betweenness	10.14	5.27	1.30
Average closeness	0.64	0.59	0.72
Average eigencentrality	0.11	0.09	0.08

components in r/cw also suggests that in comparison to r/ggw, r/cw is more disjointed and less tightly-bound as a community.

4 DISCUSSION

The conversational and network aspects highlighted above provide insight into how the gendered habitus, organizational habitus, and the emergent habitus of online community practices intersect, creating a unique perspective, goals, and values for each community. Specifically, the women-centric forums r/cw & r/ggw help bolster collective resistance and use discussions as a form of everyday resilience building - a practice similar to offline gendered contexts [26]. The anecdotal sharing of personal experiences demonstrates a strong sense of fellow feeling, expressions of peer parity, and overall commitment to the community [27]. Thus, a key underlying force in the creation of this online habitus of support and resilience is the essence of instilling empowerment through social learning in the form of peer-to-peer knowledge sharing and gradual development

of collective wisdom [8]. Maintaining community and preserving habitus norms is another key facet of such forums, which shows the keen sense of protection and community nurturing that develops in these channels [32].

Yet, the habitus in each of the forums depicts a unique form of collective purpose and mission. *r/csc* for instance, represents a community which serves as an informational hub and as such, uses a weak tie model of attachment [12]. Conversely, the other two forums demonstrate a strong sense of community, collective action, and shared understanding of the trauma and struggles women face in navigating their careers [10, 13]. In conjunction with the feminist conceptualization of technology [31], which shows how gender shapes, and is in turn shaped by technology use, we see how the construct of gender emerges in the habitus constructed in these online communities. For example, given the more informational outlook of *r/csc*, a form of gendered silencing manifesting in muting discussions around sociocultural barriers and challenges in the workplace emerges. In sharp contrast, the women-centric forums attempt to empower women and other marginalized groups, and advocate for reclaiming the equivocality and dignity across multiple contexts of expression [7, 10]. Expanding this notion of support and resilience -building for suppressed voices, we observe a further critical distinction between the nature of advocacy supported by *r/ggw* and *r/cw*. In particular, content themes indicate that *r/ggw* invokes a sharper critique of the symbolic manifestation of dominance in technical spaces which has been captured through concepts such as technochauvinism [4]. On the other hand, *r/cw* represents an alliance of professional women to network, inspire, and unite, and provides a ground for community-building and support. While a similar spirit exists in *r/ggw*, it exhibits an additional dimension by opposing the habitus of exclusion and cultural oppression through collective community endeavors and discussions. This aspect gives such online collectives a unique purpose, vision, and voice centered on empowerment and resistance.

5 CONCLUSION AND FUTURE WORK

Through a mixed-methods comparative investigation, this paper provides an initial foray into understanding how online communities may serve to help underrepresented voices develop collective resistance and resilience. These strategies can be used to subvert the tensions and restraining forces which may impede the identity formation of underrepresented groups in the IT workplace. Our findings can help provide recommendations for the design and community moderation practices of such forums to further boost their cooperative efficacy. Future work in this regard can expand the comparative axis of exploration through further consolidation of content themes and through an investigation of latent linguistic and emotional markers from a more quantitative perspective. Complementary qualitative work may further investigate the mechanisms of participation and engagement that help sustain these spaces.

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